

SSZG622 - Software Project Management

Individual Assignments 1 (10 marks) and 2 (10 marks)

Instructions:

- **Project Topic:** You are expected to **select any one of three assigned topics specific software project management** for both Assignment 1 and Assignment 2. Ensure that your report focuses on your selected topic throughout.
- Replace the placeholder text (e.g., [your project topic – X], [your name], [your id]) with your specific details.
- **Certification:** Include the certification statement as provided in the report template to confirm the originality of your work. Be aware that a similarity score above 25% will result in zero marks.
- **Submission document name:** Submit your report as a pdf file named exactly as your Student ID (all lowercase letters)
- **Assignment 1:** Draft of Executive Summary and Section 1- Real life scenario (see the Report template) for the selected topic are required for this submission. Due date is before mid-semester test. Exact date and mode of submission will be announced by Week 5.
- **Assignment 2:** Final version as specified in the Report template for the selected topic is required for this submission. Due date is before comprehensive exam. Exact date and mode of submission will be announced by Week 5.
- **Submissions:** Ensure your report is formatted clearly (e.g., 10 or 12-point font, single space), and submitted as a PDF or Word document via the course portal.

Executive Summary (Max 1 page)

Objective:

Provide a concise overview of your report, summarizing the real-life scenario, key findings from your analysis, and proposed recommendations related to your selected software project management topic (X). The Executive Summary should engage readers by highlighting the relevance of your report and encouraging them to explore the full content.

Placement:

Include the Executive Summary immediately after the Cover Page and before Section 1 (Real-life Scenario) in your report.

Guidance:

- **Purpose:** The Executive Summary serves as a snapshot of your report, allowing busy professionals to quickly grasp the main points without reading the entire document. It should be self-contained and compelling.
- **Content Structure:** Include the following elements in 3–4 brief paragraphs:
 - **Introduction to Topic and Scenario:** Briefly introduce your assigned topic (X) and the anonymized organization/project(s) where it was applied.
 - Example: If X is Agile Methodology, state, “This report examines the use of Agile Methodology in a mid-sized IT firm ABC Corp. delivering a customer portal project.”
 - **Key Findings from Analysis:** Summarize the 3–4 key weaknesses or drawbacks identified in how X was implemented, linking them to project outcomes.
 - Example: “The analysis revealed issues such as inconsistent sprint planning and lack of stakeholder engagement, leading to delays.”
 - **Recommendations Overview:** Highlight your key recommendations to address the weaknesses, emphasizing their alignment with software project management principles.
 - Example: “Recommendations include adopting structured sprint planning and regular stakeholder reviews to enhance project alignment.”
 - **Engagement Hook:** Conclude with a brief prompt to engage readers, encouraging reflection or application of your findings.
 - Example: “How can Agile practices be tailored to your workplace to avoid similar challenges?”
- **Tone and Style:**
 - Use clear, professional language suitable for a managerial audience.
 - Avoid jargon unless it’s central to your topic (e.g., “Scrum” for Agile).
- **Length:** Maximum 1 page (approximately 250–300 words). Use short paragraphs and bullet points (if needed) for readability.

Example Executive Summary:

This report explores the application of Risk Management in a mid-sized software development firm delivering a financial application. The project, involving a 10-person team, used a risk register but faced challenges due to late risk identification and inadequate mitigation strategies, resulting in a three-week delay.

Analysis revealed three key weaknesses: delayed risk identification, lack of stakeholder involvement in risk planning, and insufficient prioritization of risks. These issues, rooted in deviations from PMBOK guidelines, led to cost overruns and scope creep.

To address these, the report recommends conducting risk identification workshops at project kickoff, involving stakeholders in risk reviews, and using a risk matrix for prioritization. These solutions align with course concepts on proactive risk management and aim to improve project outcomes.

How would you adapt these recommendations to enhance risk management in your workplace? Explore the full report for detailed insights and consider Session 7's lecture on risk management for further context.

1 - Real-life Scenario (3-4 pages)

Objective:

Describe a real-life scenario from your current or previous workplace where your assigned software project management topic (X) was applied. This section should provide context, detail how X was implemented, and discuss the outcomes and challenges faced.

Guidance:

- **Company Description:** Provide a brief, anonymized overview of the organization (e.g., "a mid-sized software development firm specializing in financial applications"). Avoid using the real company name or identifiable details.
 - Example: If your topic is Risk Management, describe the company's size, industry, and typical project types (e.g., "a large IT consultancy delivering enterprise solutions").
- **Project Description:** Select 1–2 projects where your topic (X) was used. Describe the project(s) briefly, including scope, objectives, and team size.
 - Example: For Agile Methodology, describe a project where Agile (e.g., Scrum or Kanban) was used, detailing sprints, roles (e.g., Scrum Master, Product Owner), and deliverables.
- **Implementation of X:** Explain how your topic was applied in the project(s). For example:
 - If X is Estimation Techniques, describe the methods used (e.g., Planning Poker, Function Point Analysis) and who was involved.
 - If X is Stakeholder Engagement, detail how stakeholders were identified, communicated with, and involved in decision-making.
- **Outcomes:** Discuss the results of applying X. Were the projects successful? Did they meet deadlines, budgets, or quality goals? Provide measurable outcomes if possible (e.g., "delivered 10% under budget" or "reduced defects by 15%").
- **Challenges:** Highlight specific difficulties or hurdles encountered while applying X. For example:
 - For Risk Management, were risks poorly identified or mitigated?
 - For Agile, did teams struggle with adapting to frequent changes?
- **Tip for Engagement:** Reflect on your personal experience in these projects. How did the application of X affect you or your team? This will help you connect the scenario to course concepts.

Length: 3 to 4 pages (approximately 600–1000 words). Use clear headings (e.g., Company Overview, Project Details, Implementation, Outcomes, Challenges) to organize your content.

2 – Analysis (2 pages)

Objective:

Critically analyze the scenario described in Section 1, focusing on the application of your assigned topic (X). Identify and discuss 3–4 key weaknesses or drawbacks in how X was implemented, linking them to software project management principles.

Guidance:

- **Critical Analysis:** Evaluate the effectiveness of X in the scenario. Consider questions like:
 - Was X applied consistently and correctly?
 - Did the approach align with best practices from the course?
 - What gaps or inefficiencies did you observe?
- **Identify Weaknesses:** Pinpoint 3–4 specific issues in the application of X. For each weakness:
 - Describe the issue clearly (e.g., "Lack of stakeholder involvement led to misaligned project goals").
 - Explain why it was a problem, using course concepts.
 - Provide evidence from your scenario to support your analysis.
- **Examples of Weaknesses:**
 - For Agile Methodology: Lack of proper sprint planning or poor team collaboration.
 - For Risk Management: Inadequate risk identification or failure to prioritize risks.
 - For Estimation Techniques: Over-reliance on expert judgment without data-driven validation.
- **Tip for Engagement:** Use a table to summarize your 3–4 weaknesses, with columns for "Weakness," "Description," and "Impact on Project." This visual aid can make your analysis clearer.
- **Reflection Prompt:** How did these weaknesses affect the project's success? Could better application of X have changed the outcome?

Length: 2 pages (approximately 300–600 words). Use clear headings for each weakness to improve readability.

Example Table:

Weakness	Description	Impact on Project
Poor Risk Identification	Risks were documented late in the project lifecycle.	Led to unexpected delays and cost overruns.
Inadequate Stakeholder Communication	Stakeholders were not updated regularly.	Caused scope creep and misaligned expectations.

3 – Recommendations (About 1 page)

Objective:

Propose actionable recommendations to address the weaknesses identified in Section 2, ensuring they are specific to your scenario and grounded in software project management concepts.

Guidance:

- **Link to Weaknesses:** For each weakness in Section 2, provide 1–2 recommendations to improve the application of X.
 - Example: If a weakness was “poor risk identification,” recommend adopting a structured risk identification process (e.g., brainstorming sessions or SWOT analysis) early in the project.
- **Justification:** Explain why each recommendation would work, referencing course concepts or industry best practices. For example:
 - For Agile: “Implementing daily stand-ups, as recommended in Scrum, can improve team collaboration by ensuring regular communication.”
 - For Estimation: “Using historical data for estimation can increase accuracy.”
- **Specificity:** Tailor recommendations to your scenario. Avoid generic suggestions like “improve communication.” Instead, specify how (e.g., “Use a stakeholder communication plan with weekly updates”).
- **External Sources:** You may reference books, articles, or online resources to support your recommendations, but ensure they are relevant to your scenario and cited properly in the References section.
- **Tip for Engagement:** Include a brief “Implementation Plan” for one recommendation, outlining steps to apply it (e.g., “Train team on risk management tools, assign a risk owner, and review risks biweekly”).

Length: About 1 page (approximately 300 words). Use bullet points or numbered lists for clarity.

Example Recommendation:

- **Weakness:** Poor risk identification.
 - **Recommendation:** Implement a risk identification workshop at project kickoff, involving all key stakeholders, as per PMBOK guidelines.
 - **Justification:** Early identification reduces the likelihood of unforeseen issues, as seen in the scenario where late risk documentation caused delays.
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References

Instructions:

List all sources (books, articles, online documents, etc.) used in preparing your report. Use a consistent citation format (e.g., APA, MLA). If no external sources were used, state: "No external references were used."

Guidance:

- Include only credible sources relevant to your topic and scenario. For example:
 - Books: "A Guide to the Project Management Body of Knowledge (PMBOK Guide)."
 - Articles: Peer-reviewed journals or reputable websites like PMI.org.
- Avoid citing non-credible sources (e.g., random blogs or unverified websites).
- **Example Citation (APA):**
Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*(7th ed.). PMI.